COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	10/14/2026	**DATE SUBMITTED:**	10/14/2026
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ACTIVITY NO. 6.1 **Functions**

1. PROCEDURE OUTPUT

FINALS.cpp | Activity6.1.cpp

kanya > Activity6.1.cpp > ...

```cpp
#include <iostream>
using namespace std;

int add(int a, int b) {
    return a + b;
}

int subtract(int a, int b) {
    return a - b;
}

int multiply(int a, int b) {
    return a * b;
}

double divide(int a, int b) {
    return (double) a/b;
}

int main () {
    int num1, num2;

    cout << "Enter the first Integer: ";
    cin >> num1;
    cout << "Enter the second Integer ";
    cin >> num2;

    cout << "Sum: " << add(num1, num2) << endl;
    cout << "Difference: " << subtract(num1, num2) << endl;
    cout << "Product: " << multiply(num1, num2) << endl;
    cout << "Quotient: " << divide(num1, num2) << endl;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya\Activity6.1.exe
Enter the first Integer: 59
Enter the second Integer 67
Sum: 126
Difference: -8
Product: 3953
Quotient: 0.880597
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

For each of the following, create the following c++ code:

1. Create a program in C++ that will convert a user-input Farenheight to Celcius and vice versa. Use functions in the program

FINALS.cpp | Activity6.1.cpp

kanya > Activity6.1.cpp > ...

```cpp
#include <iostream>
using namespace std;

double fahrenheitToCelsius(double fahrenheit) {
    return (fahrenheit - 32) * 5.0 / 9.0;
}

double celsiusToFahrenheit(double celsius) {
    return (celsius * 9.0 / 5.0) + 32;
}

int main() {
    int choice;
    double temperature;

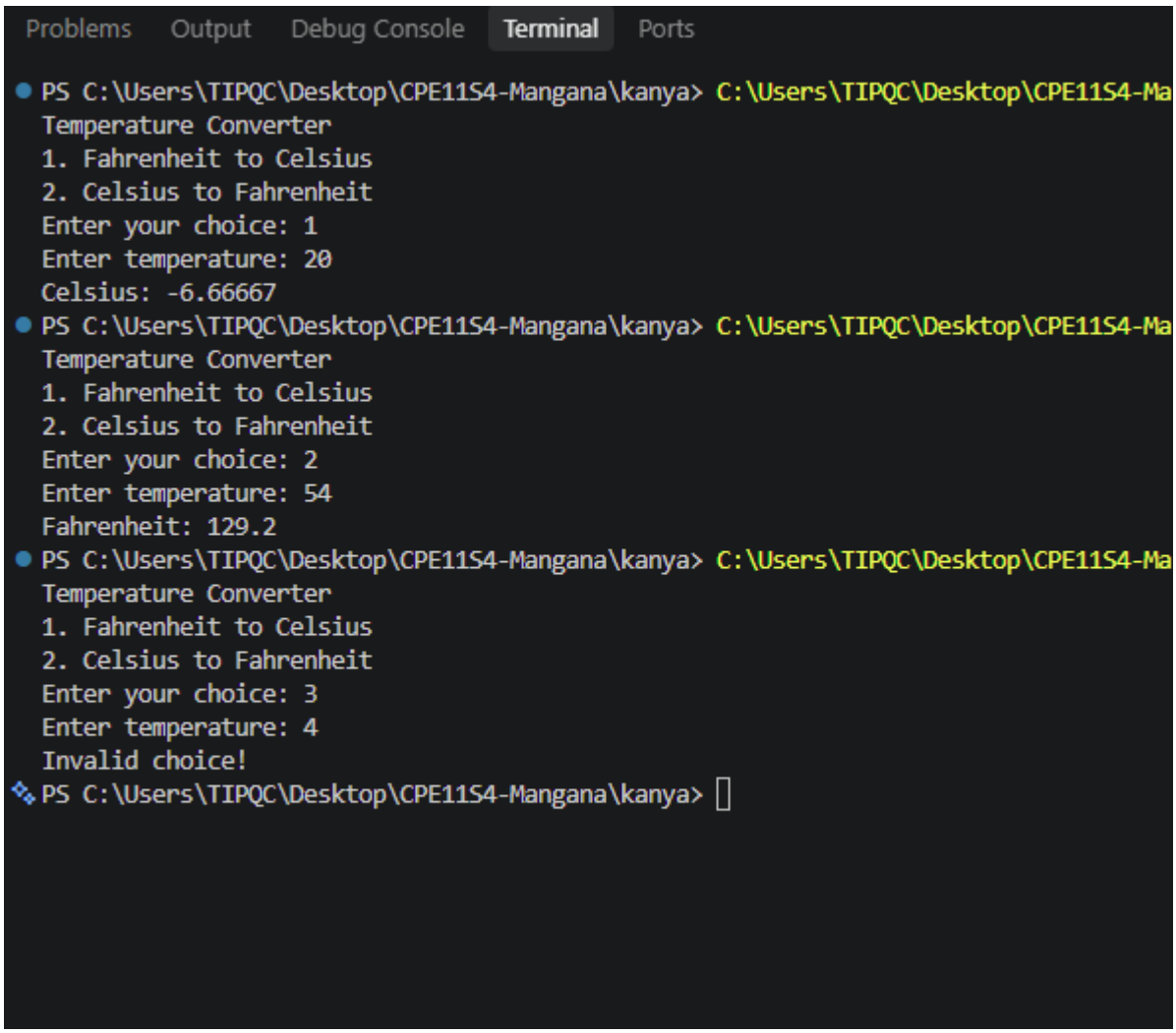
    cout << "Temperature Converter" << endl;
    cout << "1. Fahrenheit to Celsius" << endl;
    cout << "2. Celsius to Fahrenheit" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    cout << "Enter temperature: ";
    cin >> temperature;

    if (choice == 1) {
        cout << "Celsius: "
             << fahrenheitToCelsius(temperature)
             << endl;
    }
    else if (choice == 2) {
        cout << "Fahrenheit: "
             << celsiusToFahrenheit(temperature)
             << endl;
    }
    else {
        cout << "Invalid choice!" << endl;
    }

    return 0;
}

```



2. Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

FINALS.cpp Activity6.1.cpp

kanya > Activity6.1.cpp > ...

```cpp
#include <iostream>
using namespace std;

double dollarsToPesos(double dollars, double rate) {
    return dollars * rate;
}

double pesosToDollars(double pesos, double rate) {
    return pesos / rate;
}

int main() {
    int choice;
    double amount, rate;

    cout << "Currency Converter" << endl;
    cout << "Enter exchange rate (1 dollar in pesos): ";
    cin >> rate;

    if (rate <= 0) {
        cout << "Invalid exchange rate!" << endl;
        return 0;
    }

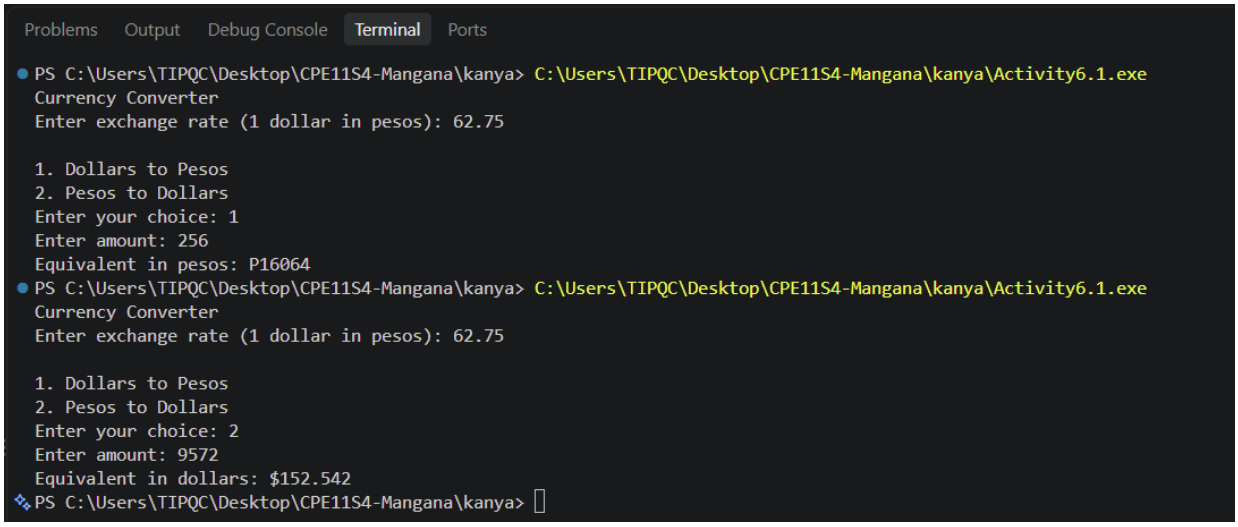
    cout << "\n1. Dollars to Pesos" << endl;
    cout << "2. Pesos to Dollars" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    cout << "Enter amount: ";
    cin >> amount;

    if (amount < 0) {
        cout << "Invalid amount!" << endl;
    }
    else if (choice == 1) {
        cout << "Equivalent in pesos: P"
             << dollarsToPesos(amount, rate)
             << endl;
    }
    else if (choice == 2) {
        cout << "Equivalent in dollars: $"
             << pesosToDollars(amount, rate)
             << endl;
    }
    else {
        cout << "Invalid choice!" << endl;
    }

    return 0;
}

```



3. Create a C++ program for a Shopping Cart Price Calculator. The program must allow the user to enter the prices and quantities of 5 different products. Store the prices and quantities in separate arrays and use for loops to process the data. Create separate functions to calculate the total cost of each product, compute the total bill, and determine the discount based on the following rules:
- 10% discount if the total bill is at least ₱3,000.
- 5% discount if the total bill is at least ₱1,500 but less than ₱3,000.
- No discount if the total bill is below ₱1,500.

Display each product's price, quantity, and subtotal, followed by the total bill, discount amount, and final amount to pay.

FINALS.cpp | Activity6.1.cpp

kanya > Activity6.1.cpp > ...

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

const int SIZE = 5;

double ProductTotal(double price, int quantity) {
    return price * quantity;
}

double TotalBill(double subtotals[], int size) {
    double total = 0;

    for (int i = 0; i < size; i++) {
        total += subtotals[i];
    }

    return total;
}

double Discount(double totalBill) {
    if (totalBill >= 3000) {
        return totalBill * 0.10;
    }
    else if (totalBill >= 1500) {
        return totalBill * 0.05;
    }
    else {
        return 0;
    }
}

int main() {
    double prices[SIZE];
    int quantities[SIZE];
    double subtotals[SIZE];

    cout << "SHOPPING CART PRICE CALCULATOR" << endl;

    for (int i = 0; i < SIZE; i++) {
        cout << "\nProduct " << i + 1 << endl;

        cout << "Enter price: P";
        cin >> prices[i];

        cout << "Enter quantity: ";
        cin >> quantities[i];

        subtotals[i] =
            ProductTotal(prices[i], quantities[i]);
    }

    double totalBill = TotalBill(subtotals, SIZE);
    double discount = Discount(totalBill);
    double finalAmount = totalBill - discount;

    cout << fixed << setprecision(2);

```

```
58
59      cout << "\n========== SHOPPING CART ==========" << endl;
60
61      cout << left << setw(12) << "Product"
62           << setw(12) << "Price"
63           << setw(12) << "Quantity"
64           << "Subtotal" << endl;
65
66      for (int i = 0; i < SIZE; i++) {
67          cout << left << setw(12) << i + 1
68               << setw(12) << prices[i]
69               << setw(12) << quantities[i]
70               << subtotals[i] << endl;
71      }
72
73      cout << "-----------------------------------" << endl;
74      cout << "Total Bill: P" << totalBill << endl;
75      cout << "Discount: P" << discount << endl;
76      cout << "Final Amount: P" << finalAmount << endl;
77
78      return 0;
79  }
80
```

OUTPUT

10% DISCOUNT

```
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya\Activity6.1.exe
SHOPPING CART PRICE CALCULATOR

Product 1
Enter price: P500
Enter quantity: 2

Product 2
Enter price: P700
Enter quantity: 1

Product 3
Enter price: P56
Enter quantity: 3

Product 4
Enter price: P67
Enter quantity: 42

Product 5
Enter price: P1000
Enter quantity: 2

========== SHOPPING CART ==========
Product     Price       Quantity    Subtotal
1           500.00      2           1000.00
2           700.00      1           700.00
3           56.00       3           168.00
4           67.00       42          2814.00
5           1000.00     2           2000.00
-----------------------------------
Total Bill: P6682.00
Discount: P668.20
Final Amount: P6013.80
```

5% DISCOUNT

```
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya\Activity6.1.exe
SHOPPING CART PRICE CALCULATOR

Product 1
Enter price: P300
Enter quantity: 2

Product 2
Enter price: P400
Enter quantity: 1

Product 3
Enter price: P250
Enter quantity: 1

Product 4
Enter price: P150
Enter quantity: 3

Product 5
Enter price: P356
Enter quantity: 1

========== SHOPPING CART ==========
Product     Price       Quantity    Subtotal
1           300.00      2           600.00
2           400.00      1           400.00
3           250.00      1           250.00
4           150.00      3           450.00
5           356.00      1           356.00
-----------------------------------
Total Bill: P2056.00
Discount: P102.80
Final Amount: P1953.20
```

NO DISCOUNT

```
PS C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya> C:\Users\TIPQC\Desktop\CPE11S4-Mangana\kanya\Activity6.1.exe
SHOPPING CART PRICE CALCULATOR

Product 1
Enter price: P100
Enter quantity: 3

Product 2
Enter price: P56
Enter quantity: 4

Product 3
Enter price: P96
Enter quantity: 4

Product 4
Enter price: P100
Enter quantity: 2

Product 5
Enter price: P12
Enter quantity: 4

========== SHOPPING CART ==========
Product     Price       Quantity    Subtotal
1           100.00      3           300.00
2           56.00       4           224.00
3           96.00       4           384.00
4           100.00      2           200.00
5           12.00       4           48.00
-----------------------------------
Total Bill: P1156.00
Discount: P0.00
Final Amount: P1156.00
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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